Mathematical Background

• Vectors and Scalars

A vector quantity has magnitude and direction (For example: Electric Field Vector, $\boldsymbol{E}$).
A scalar quantity has magnitude, only (For example: Electric Charge, q).

Vectors and Scalars can be multi-valued functions. For example

$$\vec{\boldsymbol{E}} = \vec{\boldsymbol{E}}(x,y,z) \quad , \quad q = q(x,y,z)$$

• Gradient, Divergence and Rotational Operators

Let's define Nabla (∇f) operator as[1]: $\nabla = \frac{\partial}{\partial x}\vec{\boldsymbol{x}} + \frac{\partial}{\partial y}\vec{\boldsymbol{y}} + \frac{\partial}{\partial z}\vec{\boldsymbol{z}}$

Nabla operator is a vector. //

• Gradient (∇f) must be applied to scalars, only. If $V = V(x,y,z)$ is a scalar, then $gradV = \nabla V$ is

$$\nabla V(x,y,z) = \frac{\partial V(x,y,z)}{\partial x}\vec{\boldsymbol{x}} + \frac{\partial V(x,y,z)}{\partial y}\vec{\boldsymbol{y}} + \frac{\partial V(x,y,z)}{\partial z}\vec{\boldsymbol{z}}$$

Scalar → ∇V → Vector

∇V can be evaluated as maximum spatial rate of change and the result of ∇V must be a vector.

• Divergence ($\nabla \cdot \boldsymbol{A}$) must be applied to vectors, only. If $\vec{\boldsymbol{A}} = A_x(x,y,z)\vec{\boldsymbol{x}} + A_y(x,y,z)\vec{\boldsymbol{y}} + A_z(x,y,z)\vec{\boldsymbol{z}}$ is a vector, then $div\vec{\boldsymbol{A}} = \nabla \cdot \vec{\boldsymbol{A}}$ is

$$\nabla \cdot \vec{\boldsymbol{A}}(x,y,z) = \frac{\partial A_x(x,y,z)}{\partial x} + \frac{\partial A_y(x,y,z)}{\partial y} + \frac{\partial A_z(x,y,z)}{\partial z}$$

Vector → $\nabla \cdot \vec{\boldsymbol{A}}$ → Scalar

$\nabla \cdot \vec{\boldsymbol{A}}$ can be evaluated as flux which is flowing outwardly and the result of $\nabla \cdot \vec{\boldsymbol{A}}$ must be a scalar.

• Rotational ($\nabla \times \boldsymbol{A}$) must be applied to vectors, only. If $\vec{\boldsymbol{A}} = A_x(x,y,z)\vec{\boldsymbol{x}} + A_y(x,y,z)\vec{\boldsymbol{y}} + A_z(x,y,z)\vec{\boldsymbol{z}}$ is a vector, then $rot\vec{\boldsymbol{A}} = \nabla \times \vec{\boldsymbol{A}}$ is

$$\nabla \times \vec{\boldsymbol{A}}(x,y,z) = \begin{vmatrix} \vec{\boldsymbol{x}} & \vec{\boldsymbol{y}} & \vec{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \left(\frac{\partial}{\partial y}A_z - \frac{\partial}{\partial z}A_y\right)\vec{\boldsymbol{x}} - \left(\frac{\partial}{\partial x}A_z - \frac{\partial}{\partial z}A_x\right)\vec{\boldsymbol{y}} + \left(\frac{\partial}{\partial x}A_z - \frac{\partial}{\partial z}A_x\right)\vec{\boldsymbol{z}}$$

Vector → $\nabla \times \vec{\boldsymbol{A}}$ → Vector

$\nabla \times \vec{\boldsymbol{A}}$ can be evaluated as maximum net rotation in the unit area and the result of $\nabla \times \boldsymbol{A}$ must be a vector.

[1] Sahoo S., (2008), Inverse vector operators, (arXiv:0804.2239)

Dr. Serkan AKSOY

• **Useful Vector Relations**

$$\vec{A}\times(\vec{B}\times\vec{C})=\vec{B}\cdot(\vec{A}\cdot\vec{C})-\vec{C}\cdot(\vec{A}\cdot\vec{B})$$

$$\vec{A}\times\vec{B}=-\vec{B}\times\vec{A}$$

• **Useful Relations for Gradient, Divergence and Rotational Operators**

✓ $\nabla\cdot(\nabla V)=\Delta V$, Δ : Laplacian

$$\nabla\cdot\nabla=\left(\frac{\partial}{\partial x}\vec{x}+\frac{\partial}{\partial y}\vec{y}+\frac{\partial}{\partial z}\vec{z}\right)\cdot\left(\frac{\partial}{\partial x}\vec{x}+\frac{\partial}{\partial y}\vec{y}+\frac{\partial}{\partial z}\vec{z}\right)=\overbrace{\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}}^{\Delta}$$

✓ $\nabla\times\overbrace{(\nabla V)}^{\vec{A}}=0 \quad\Rightarrow\quad \vec{A}=\nabla f \quad\Rightarrow\quad \nabla\times\vec{A}=0$

✓ $\nabla\cdot\overbrace{(\nabla\times\vec{A})}^{\vec{B}}=0 \quad\Rightarrow\quad \vec{B}=\nabla\times\vec{A} \quad\Rightarrow\quad \nabla\cdot\vec{B}=0$

✓ $\nabla\times(\nabla\times\vec{A})=\nabla(\nabla\cdot\vec{A})-\Delta\vec{A}$

✓ $\nabla\times(a(\vec{r})\cdot\vec{A})=a(\vec{r})\cdot\nabla\times\vec{A}+\nabla a(\vec{r})\times\vec{A}=a(\vec{r})\cdot\nabla\times\vec{A}+\vec{A}\times\nabla(-a(\vec{r}))$

✓ $\nabla(\vec{A}\cdot\vec{B})=(\vec{A}\cdot\nabla)\vec{B}+(\vec{B}\cdot\nabla)\vec{A}+\vec{A}\times\nabla\times\vec{B}+\vec{B}\times\nabla\times\vec{A}$

✓ $\nabla\cdot(a(\vec{r})\cdot\vec{A})=a(\vec{r})\cdot\nabla\cdot\vec{A}+\nabla a(\vec{r})\cdot\vec{A}$

✓ $\nabla\cdot(a(\vec{r})\cdot\nabla b(\vec{r}))=a(\vec{r})\cdot\Delta b(\vec{r})+\nabla a(\vec{r})\cdot\nabla b(\vec{r})$

✓ $\nabla e^{i\vec{k}\vec{r}}=i\vec{k}e^{i\vec{k}\vec{r}}$

✓ $\nabla\cdot\left(\vec{A}e^{+i\vec{k}\vec{r}}\right)=i\vec{k}\cdot\vec{A}e^{i\vec{k}\vec{r}}$

✓ $\nabla\times\left(\vec{A}e^{i\vec{k}\vec{r}}\right)=i(\vec{k}\times\vec{A})e^{i\vec{k}\vec{r}}$

where $\vec{A}$ is a constant vector. $\vec{A}$ and $\vec{k}$ may also be complex.

• **Divergence Theorems** ($\overrightarrow{ds}=\vec{n}\cdot ds$)

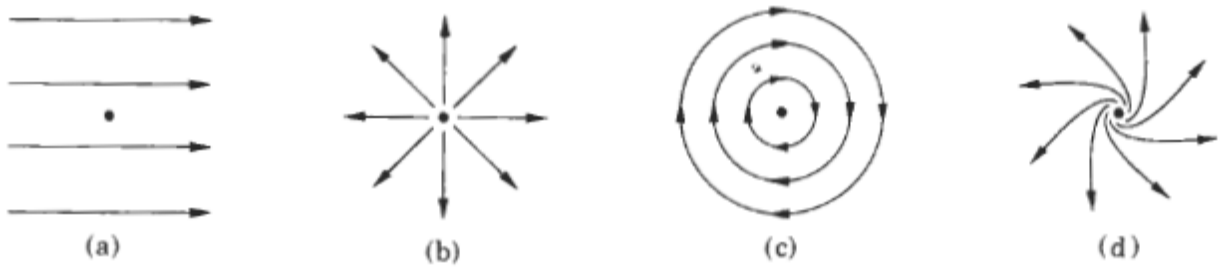
$$\underbrace{\iiint_v \nabla\cdot\vec{A}dv}_{small-scale}=\underbrace{\iint_S \vec{A}\cdot\overrightarrow{ds}}_{large-scale}$$

• **Stokes Theorems**

$$\underbrace{\iint_S \nabla\times\vec{A}\cdot\overrightarrow{ds}}_{small-scale}=\underbrace{\int_l \vec{A}\cdot\overrightarrow{dl}}_{large-scale}$$

Dr. Serkan AKSOY

• **Vector Field Lines** [Jefimenko, 1989].

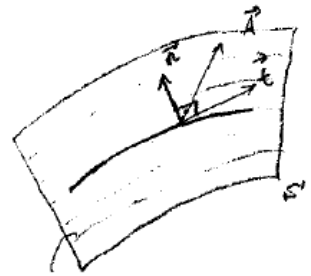


Field lines of a vector field **V** near a point where

(a) div $\mathbf{V} = 0$, curl $\mathbf{V} = 0$
(b) div $\mathbf{V} \neq 0$, curl $\mathbf{V} = 0$
(c) div $\mathbf{V} = 0$, curl $\mathbf{V} \neq 0$
(d) div $\mathbf{V} \neq 0$, curl $\mathbf{V} \neq 0$.

• **Normal and Tangential Vector Concept**

Let's image a surface S. The normal ($\vec{n}$) ve tangential ($\vec{t}$) vectors satisf following equalities,



$f(x,y,z) = \alpha_0$ (constant)

$grad f \perp \vec{t}$

$grad f \parallel \vec{n}$

$\vec{n} \cdot \vec{t} = 0 \Rightarrow$ They are perpendicular to each other.

$\vec{n}$ and $\vec{t}$ are unity vectors mean that $|\vec{n}| = 1$, $|\vec{t}| = 1$

Any vector on the surface S can be written as

$$\vec{A} = A_n \vec{n} + A_t \vec{t}$$

• **COORDINATE SYSTEMS**

• Cartesian (Rectangular) Coordinate System :

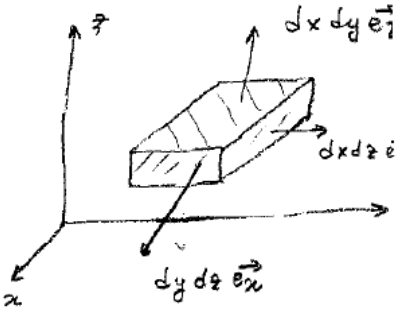
$f = f(x,y,z)$

$\vec{A} = A_x \vec{e_x} + A_y \vec{e_y} + A_z \vec{e_z}$

* $ds_x = dx\,dy$, $ds_y = dx\,dz$, $ds_z = dy\,dz$

* $dV = dx\,dy\,dz$

$d\vec{\ell} = dx\,\vec{e_x} + dy\,\vec{e_y} + dz\,\vec{e_z}$; $|\vec{e_x}| = |\vec{e_y}| = |\vec{e_z}| = 1$ Unity vectors



- Cylindrical Coordinate System:

$$f = f(\rho, \varphi, z)$$

$$\vec{A} = A_\rho \vec{e}_\rho + A_\varphi \vec{e}_\varphi + A_z \vec{e}_z$$

$$d\vec{l} = dr\vec{e}_r + \rho d\varphi \vec{e}_\varphi + dz\vec{e}_z$$

$$* \quad ds_r = \rho d\varphi dz \, \vec{e}_r$$

$$ds_\varphi = d\rho dz \, \vec{e}_\varphi$$

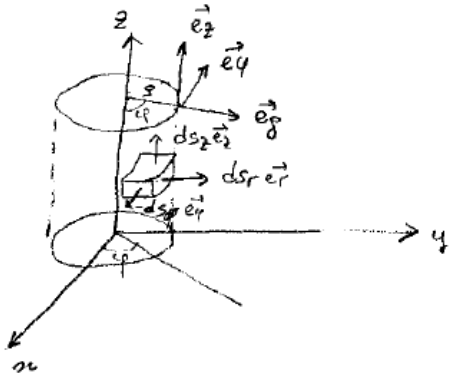
$$ds_z = \rho d\rho d\varphi \, \vec{e}_z$$

$$* \quad dv = \rho d\rho d\varphi dz$$

$$\vec{e}_\rho = \cos\varphi \vec{e}_x + \sin\varphi \vec{e}_y$$

$$\vec{e}_\varphi = -\sin\varphi \vec{e}_x + \cos\varphi \vec{e}_y$$

$$\vec{e}_z = \vec{e}_z$$



- Spherical Coordinate System:

$$f = f(\rho, \varphi, \theta)$$

$$\vec{A} = A_\rho \vec{e}_\rho + A_\varphi \vec{e}_\varphi + A_\theta \vec{e}_\theta$$

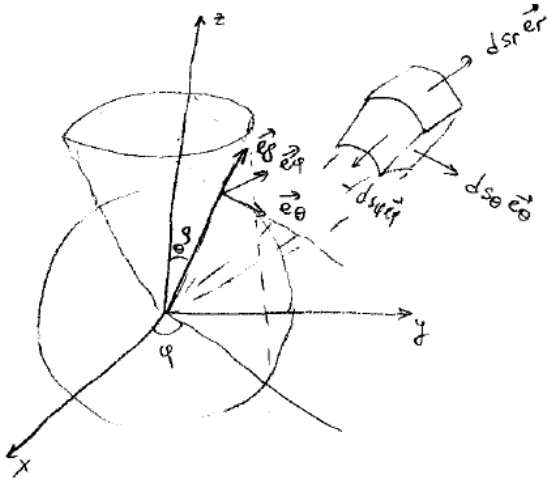
$$d\vec{l} = d\rho \vec{e}_\rho + \rho d\theta \, \vec{e}_\theta + \rho \sin\theta d\varphi \vec{e}_\varphi$$

$$* \quad ds_r = \rho^2 \sin\theta \, d\theta d\varphi$$

$$ds_\theta = \rho \sin\theta \, d\rho d\varphi$$

$$ds_\varphi = \rho d\rho d\theta$$

$$* \quad dv = \rho^2 \sin\theta d\rho d\theta d\varphi$$



$$\vec{e}_\rho = \sin\theta\cos\varphi \vec{e}_x + \sin\theta\sin\varphi \vec{e}_y + \cos\theta \vec{e}_z$$

$$\vec{e}_\theta = \cos\theta\cos\varphi \vec{e}_x + \cos\theta\sin\varphi \vec{e}_y - \sin\theta \vec{e}_z$$

$$\vec{e}_\varphi = -\sin\varphi \vec{e}_x + \cos\varphi \vec{e}_y$$

$$\int_e \vec{A} d\vec{l} \quad , \quad \int_s \vec{A} d\vec{s} \quad \int_v f \, dv$$

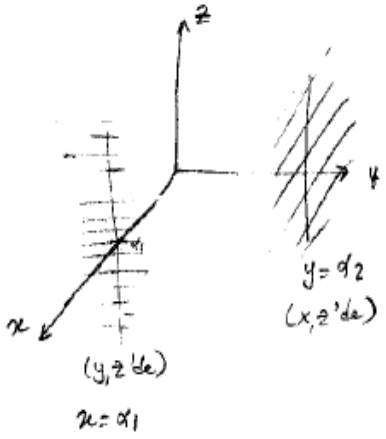
• ÇOK DEĞİŞKENLİ FONKSİYONLAR ve YÜZEYLER

$f(x,y,z)$ gözönüne alınsın

$x \in (-\infty, +\infty)$
$y \in (-\infty, +\infty)$
$z \in (-\infty, +\infty)$

$f(x_0,y_0,z_0) = \alpha_0$ değerini alırki bu $f(x,y,z) = \alpha_0$ (sabit) olan x,y,z değerleri bir yüzeyi tan.

nınlar. $f(x,y,z) = ax+by = \alpha_0 \Rightarrow$ bir düzlemi (x,y'de)
$f(x,y,z) = x = \alpha_0 \rightarrow$ " " (x'de)



$x^2+y^2+z^2 = \alpha \rightarrow$ küre
$x^2+y^2 = b \rightarrow$ dairesel silindir.

• EĞRİLER ve PARAMETRİK GÖSTERİLİM

$x = \alpha(t)$
$y = \beta(t)$
$z = \gamma(t)$
} t herhangi bir parametre t değiştikçe bir eğri gösterirler.

$f(x,y,z) = f(\alpha(t), \beta(t), \gamma(t))$

Eğrinin parametrik denklemleri



$\frac{\overrightarrow{PP'}}{\Delta t}$'nin bileşenleri $\left(\underbrace{\frac{\alpha(t+\Delta t)-\alpha(t)}{\Delta t}}_{x}, \underbrace{\frac{\beta(t+\Delta t)-\beta(t)}{\Delta t}}_{y}, \underbrace{\frac{\gamma(t+\Delta t)-\gamma(t)}{\Delta t}}_{z}\right)$

yay uzunluğu s olsun

$ds^2 = dx^2 + dy^2 + dz^2$

$\Delta t \to 0$ olursa, $\overrightarrow{PP'}$ eğriye değet vektör olur. ($\vec{t}$ ile gösterilir.) (P P'ye yaklaşıyor). $\vec{t}$'nün bileşenleri → kartezyende

$\vec{t} \cdot \vec{t} = \left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2$ $\quad \vec{t}(\alpha'(t), \beta'(t), \gamma'(t))$ olur. $\vec{t}$'nin yönünde t artıkça

$= 1 \Rightarrow |\vec{t}| = 1$ → birim vektör ← ilerlemeyle yön dür. Kartezyende $\alpha'(t)\vec{e_x} + \beta'(t)\vec{e_y} + \gamma'(t)\vec{e_z}$

Örnek: $x = a\cos\phi$
$y = a\sin\phi$
$z = 0$, $\phi \in (0, 2\pi) \Rightarrow$ değet vektör koordinatlarını bulunuz.



$x^2+y^2 = a^2$ bir daire ($z=a$ yarısında $z=a$' düzleminde)
$\vec{t} = \vec{t}(-a\sin\phi, a\cos\phi, 0)$ olur.

2.1) Eğri boyunca türev alınırsa

$f(x,y,z) = f(x(t), y(t), z(t)) = F(t)$ ②

$$\frac{dF(t)}{dt} = F'(t) = \frac{\partial f}{\partial \alpha}\cdot\frac{d\alpha}{dt} + \frac{\partial f}{\partial \beta}\frac{d\beta}{dt} + \frac{\partial f}{\partial \gamma}\frac{d\gamma}{dt} = \left(\frac{\partial f}{\partial x}\vec{e}_x + \frac{\partial f}{\partial y}\vec{e}_y + \frac{\partial f}{\partial z}\vec{e}_z\right)\left(\frac{d\alpha}{dt}\vec{e}_y + \frac{d\beta}{dt}\vec{e}_y + \frac{d\gamma}{dt}\vec{e}_z\right)$$

gradyant

Teğet vektör bileşenleri

$\Rightarrow \text{grad}f \cdot \vec{t} = F'(t)$ Bir eğri boyunda alınan türev.

[SKALER] → [grad f] → [VEKTÖREL]

2.2) Yüzey boyunca türev

- $F(t) = f(x,y,z) = f(\alpha(t), \beta(t), \gamma(t)) = \alpha_0 \Rightarrow$ yüzey olur. Türev



$\frac{dF(t)}{dt} = 0 = \text{grad}f \cdot \vec{t}$ olacaktır.

Birbirine dikdir. $\text{grad}f \perp \vec{t}$

$\text{grad}f \parallel \vec{n}$

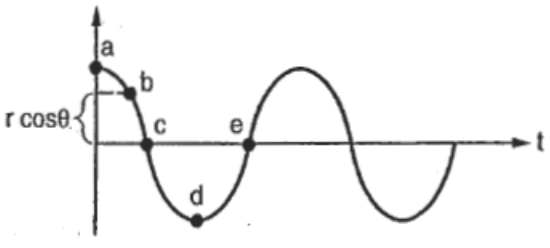
$\vec{n} = \frac{\text{grad}f}{|\text{grad}f|}$ → grad f'nin birim vektörüdür.

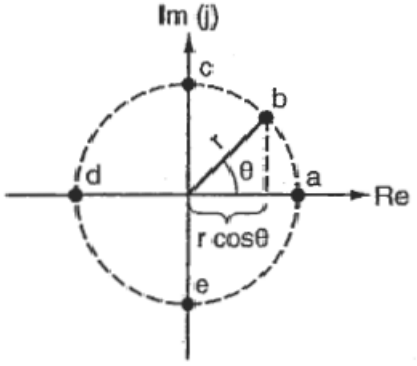
- **PHASOR REPRESENTATION**

$$re^{j\theta} = r\cos\theta + jr\sin\theta$$

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$re^{j\theta} = r\angle\theta$$





real part of the phasor is found as $\text{Re}(re^{j\theta}) = r\cos\theta$

imaginary part is $\text{Im}(re^{j\theta}) = r\sin\theta$.

• A WELL-POSED PROBLEM

An initial boundary value problem is said to be well-posed if

- it has a solution.
- the solution is unique.
- the solution depends continuously on source term and initial and boundary data.

The well-posed problem means that small changes in source term and initial and boundary data produce small change in the solution.

Dr. Serkan AKSOY